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HYBRID NUMERICAL METHOD FOR THE EVALUATION OF THE SEISMIC PROTECTION OF BUILDINGS BASED ON DIGITAL TWINS

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We present the results of numerical and experimental investigations of vibration of a residential building and its seismic protection by means of rubber isolators made in Ukraine from natural rubber. The numerical studies of vertical oscillations of the building with rubber supports on the level of reinforced-concrete grid were performed with the help of dynamic digital twins developed on the basis of the finite-element method. The numerical studies of the vibrations of digital twins were carried out according to the records of soil vibration accelerations obtained under the effects of railway trains, as well as for the effects determined according to the accelerograms of earthquakes. The numerical analysis of buildings on seismic supports with regard for the seismic loads was performed with evaluation of the factors of safety against overturning of the residential building. We show that the solution to the problem of vibration and seismic protection of buildings is possible in the case of application of the rubber elements as isolators characterized by a nonlinear dependence of stiffness on the load.

Keywords: hybrid numerical method, digital twins, dynamic and nondestructive surveys, rubber isolators, vibration-isolated grid, Vrancea earthquake zone.

Introduction

As a specific feature of buildings constructed near railways, we can mention the fact that the seismicity of construction site design may reach 7 points according to the scales of seismic intensity. For the protection of buildings against seismic impacts, the stiffness of seismic supports in the vertical direction should be much higher than in the horizontal direction. The estimation of stiffness is based on the analysis of the enforced vibrations of buildings and structures as multi-mass systems modeled by using the equations of motion. The equations of motion are obtained by using the D'Alembert and Hamilton principles or Lagrange equations (see, e.g., [17, 28]). The dynamics of systems with multiple degrees of freedom and the application of possible displacements principles and Hamilton's principle for deducing the equations of motion of linear and nonlinear dissipative systems were considered in [1, 6, 12, 25]. In [9, 29], the results on the experimental and theoretical evaluation of the dynamic effects of railway transport on soil, foundations, and building structures were discussed.

The methods used for the analysis of the response of buildings and structures to seismic actions are presented in [7, 8, 11, 14, 16, 17, 22–24, 26–28, 30–34]. The structures of seismic supports and methods used for the

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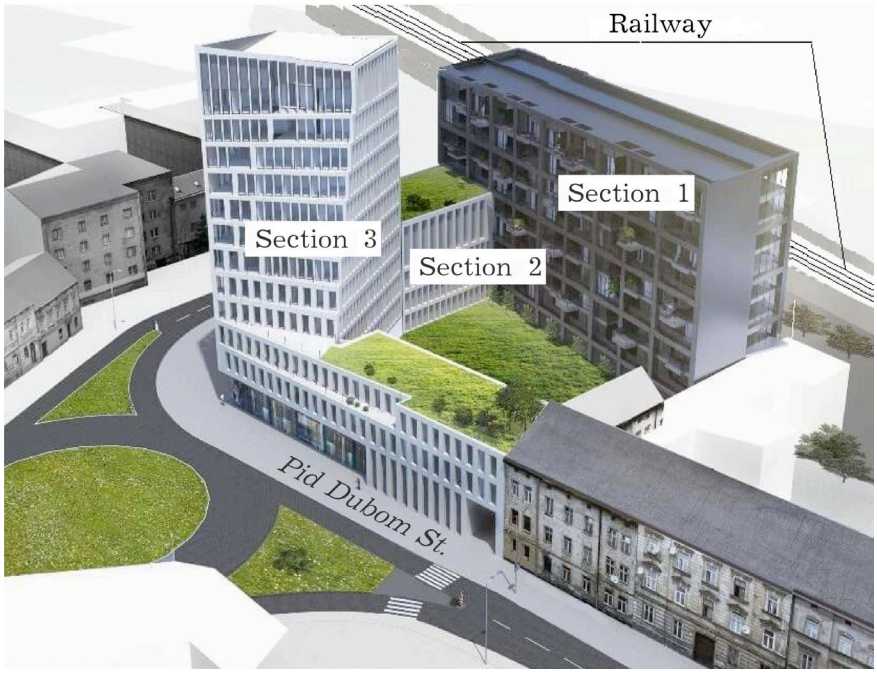


Fig. 1. The situation plan of the construction site for the residential complex at 26 Pid Dubom St. (PDRC) in the city of Lviv.evaluation of their parameters and application in the design of seismic protection systems of the buildings are listed in [10, 18, 23].

The present work is devoted to modeling of the "Pid Dubom" residential complex (PDRC) response (Fig. 1) to the dynamic effects of railway transport in the Vrancea earthquake-prone zone with the help of digital twins and the Internet of Things (IoT) application. The construction site in the city of Lviv is located near the railway track. Prior to the design of the PDRC buildings, it was experimentally determined that the vibration levels of the floors in the buildings residential premises exceeded the values permissible according to the sanitary standards for the dynamic effects of passenger and freight trains. First of all, the dynamic effects of trains cause the elevated (close to resonant) vertical vibrations of the floors in the buildings located along the railway track. To protect the residential buildings against the vertical and horizontal vibrations propagating through the ground, a vibroisolation should be arranged on the foundation level by using rubber vibration supports. The efficiency of the building floors vibroisolation is ensured by the 3–4 times range ratio of the frequency of forced impacts from the trains and the own vibrations frequency of the building on vibration supports. This condition is ensured if we use vibration supports made of low-modulus rubber based on natural rubber.

According to the Eurocode 8 recommendations [5], when designing an earthquake protection system with layered rubber-metal seismic supports, it is necessary to accept the vertical stiffness of seismic isolators by 150 times higher than the horizontal stiffness. The vertical vibrations period for a building on seismic supports should be no more than 0.1 s. However, such a requirement does not ensure protecting against the railway transport dynamic effects. For the effective vibration protection of buildings against the railway influences, the vertical vibrations period exceeding 0.2 s should be accepted for the buildings on vibration supports. The solution of this problem is achieved by the usage of the rubber isolators with the vertical stiffness exceeding the horizontal one by no more than 5–7 times. According to the results of the tests of rubber isolators conducted in Ukraine, it is found that the proposed rubber isolators meet the foregoing requirement as to the vertical and horizontal stiffness ratio.

The seismic and vibration protection method proposed and patented in Ukraine [10] is intended both to increase the seismic resistance of building structures and to provide the comfortable living conditions in buildings located near

the railway tracks. The protection system with rubber isolators ensures a reduction of seismic loads on the building structures by 1.5 to 2 times. The use of seismic protection is governed by the European and national standards EN 1998 1:2004 Eurocode 8 [5] and DBN V.1.1-12:2014 (State Building Normative) [3].

The objective of this study is to justify, by means of digital twins and IoT, the use of rubber isolators for the protection against both the railway transport dynamic effects and the seismic loads.

During the mathematical modeling of stress-strain state for PDRC (Fig. 1), several types of difficulties were encountered.

At the initial stage a digital twin, i.e., a digital model was constructed taking into account the changes made during the construction of buildings of the complex in comparison with the original project including the structural elements replacements, the actual grade of concrete used during the complex renovation (for instance, during the non-destructive testing of a building partially destroyed as a result of a rocket strike at 6-A Lobanovskyi Avenue in Kyiv [15], some discrepancies were found between the design and actual grades of its structural elements concrete), real physical and mechanical properties of building materials, real properties of soils, final geometric dimensions of buildings, etc. Those problems required the detailed on-site surveys of the complex at 26 Pid Dubom St. with the use of the geometrical analysis, non-destructive tests and other methods. All the works were carried out by the co-authors of the paper during the construction of buildings of the complex and after its completion and commissioning.

The results of the experimental studies directly affect both the correctness of the initial determination of boundary conditions and the correctness of the complex finite element model.

The boundary conditions, in addition to the complex geometric characteristics, include the physical and mechanical characteristics of all structural elements of the complex. The initial conditions include the accelerograms (which take into account the real seismicity of construction site at the complex location in Vrancea area and the dynamic effects due to the trains and vehicles movement) experimentally obtained for the direct dynamic calculation of the complex digital twin. Therefore, in addition to the mathematical modeling, the field studies of the soil and buildings floors vibration in the Pid Dubom residential complex built at a distance of 20–30 m from the Kyiv–Lviv railway track should be considered in the paper. Such tasks usually arise when the buildings are erected on the construction sites located in the earthquake-prone zones near the railway tracks. Here, the seismicity of the construction site is 7 points according to the European and Ukrainian scales of seismic intensity [3].

As a result, for modeling the stress-strain state of buildings and structures subject to dynamic effects, it is necessary to develop such hybrid systems, which would allow researches using IoT. To study the strength and vibration characteristics of the complex buildings and soil base, in addition to the traditional packages of LIRA 9.4 or SCAD application software, the IoT systems of sensors for non-destructive testing are necessary. Such IoT system is combined into one information network by means of the cloud technologies and allows obtaining in online mode the information necessary for revealing the particular buildings features. This is the second stage of the digital twin construction for PDRC.

1. Equation of Motion for a Flat Model of the Building under Dynamic Effects

The study of the gradual and angular vibrations of the multi-mass nonlinear dynamic models with seismic protection systems is performed using the step-by-step integration of differential nonlinear equations (direct dynamic method) for the specified accelerograms (kinematic impact at the building base during earthquakes). The numerical studies of the forced vibrations in the “pile base-seismic supports-foundation-building” system under seismic effects determined by the accelerograms at the building base level were carried out using the multi-mass flat (Fig. 2) computational dynamic models (CDM).

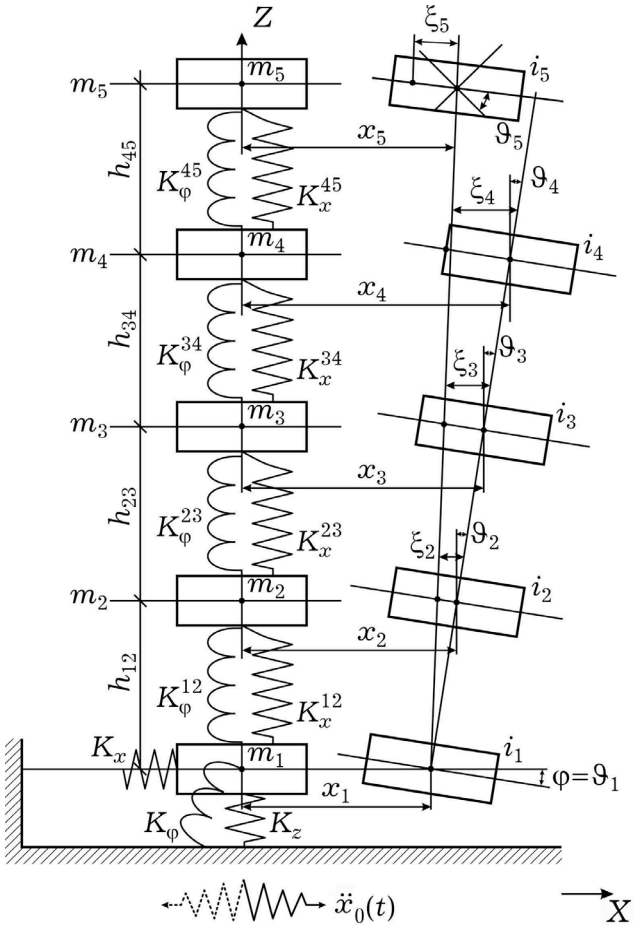


Fig. 2. The building flat CDM with regard for the base and seismic isolation stiffnesses (K_x, K_z, K_φ), mass of the foundation m_1 and moment of inertia i_1, floor structures masses $m_2,\ldots,m_5$ and moments of inertia $i_2,\ldots,i_5$, as well as the shear $K_x^{12},\ldots,K_x^{45}$ and angular $K_\varphi^{12},\ldots,K_\varphi^{45}$ stiffnesses of building structures under seismic impacts $x_0(t)$.

Considering the damping according to the Voigt hypothesis and dry friction forces at the foundation level, the forced vibrations differential equations in a matrix form for a building flat multi-mass CDM are obtained in the following form:

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{K}(\mathbf{x})\mathbf{x}(t) + \mathbf{H}\,\mathrm{sgn}(\mathbf{x}(t)) = -\mathbf{M}\ddot{\mathbf{x}}_0(t). \tag{1}$$

where $\mathbf{M}$, $\mathbf{C}$, $\mathbf{K}(\mathbf{x})$, and $\mathbf{H}$ are the matrices of the system masses and moments of inertia, damping, nonlinear stiffness and dry friction forces, respectively; $\ddot{\mathbf{x}}(t)$, $\dot{\mathbf{x}}(t)$, $\mathbf{x}(t)$ are the vectors of acceleration, velocity, and displacement of the building foundation and storey floors; $\ddot{\mathbf{x}}_0(t)$ is the soil acceleration vector (the accelerograms recorded during an earthquake or synthesized) at the building base.

During the soil base horizontal vibrations $\mathbf{x}_0(t)$, the building foundation (mass m_1) displaces in the horizontal direction x_1 and rotates around the horizontal axis by angle φ (Fig. 2). The masses of building stories m_j gradually displace relative to the foundation (coordinates ξ_j) and rotate around the horizontal axis (coordinates ϑ_j), $j=1,2,\ldots,5$, see Fig. 2. Thus, the vectors of generalized coordinates without taking into account

[9] and with taking into account the moments of inertia of the stories masses are as follow:

$$\mathbf{q}^{\mathrm{T}} = (x_1, \xi_2, \xi_3, \xi_4, \xi_5, \varphi), \qquad \mathbf{W}^{\mathrm{T}} = (x_1, \xi_2, \xi_3, \xi_4, \xi_5, \varphi, \vartheta_2, \vartheta_3, \vartheta_4, \vartheta_5).$$

To determine the absolute displacements and accelerations of the building foundation and stories, the absolute coordinates should be used:

$$\mathbf{q}^{\mathrm{T}} = (x_1, x_2, x_3, x_4, x_5, \varphi, \vartheta_2, \vartheta_3, \vartheta_4, \vartheta_5).$$

The equation of the CDM motion is obtained by using the Lagrange equations of the second kind [16, 20]:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) + \frac{\partial}{\partial q_i}(U - T) = 0,$$

where q_i are the generalized coordinates ($q_m = x_m$, $m = 1, 2, \ldots, 5$, $q_6 = \varphi$, $q_{n+5} = \vartheta_n$, $n = 2, \ldots, 5$), $\dot{q}_i$ are the generalized velocities, T is the kinetic energy of the system:

$$T = \frac{1}{2}\Big(m_1\dot{x}_1^2 + m_2\dot{x}_2^2 + m_3\dot{x}_3^2 + m_4\dot{x}_4^2 + m_5\dot{x}_5^2 + i_{\text{total}}\dot{\varphi}^2 + i_2\dot{\vartheta}_2^2 + i_3\dot{\vartheta}_3^2 + i_4\dot{\vartheta}_4^2 + i_5\dot{\vartheta}_5^2\Big),$$

and U is potential energy of the system:

$$U = \frac{1}{2}\Big(\Big(K_x x_1^2 + K_{12}(x_2 - x_1 - h_{12}\varphi)^2 + K_{23}(x_3 - x_2 - h_{23}\varphi)^2 + K_{34}(x_4 - x_3 - h_{34}\varphi)^2\Big) + K_{45}(x_5 - x_4 - h_{45}\varphi)^2 + K_\varphi\varphi^2 + K_\varphi^{12}(\vartheta_2 - \varphi)^2 + K_\varphi^{23}(\vartheta_3 - \vartheta_2 - \varphi)^2 + K_\varphi^{34}(\vartheta_4 - \vartheta_3 - \varphi)^2 + K_\varphi^{45}(\vartheta_5 - \vartheta_4 - \varphi)^2\Big).$$

Here, $i_{\text{total}} = I_{\text{IM}} + \sum_j m_j h_{j,\text{IM}}^2$, I_{IM} is the moment of inertia of the building relative to the horizontal axis passing through the center of gravity; $h_{j,\text{IM}}$ is the distance from the center of gravity to the jth mass of the building, and K_x, K_{12} K_{23}, K_{34}, K_{45}, K_φ, K_φ^{12}, K_φ^{23}, K_φ^{34}, K_φ^{45} are described in the legend of Fig. 2.

In the expanded form, the system of differential equations (1) (10 degrees of freedom) can be written as follows:

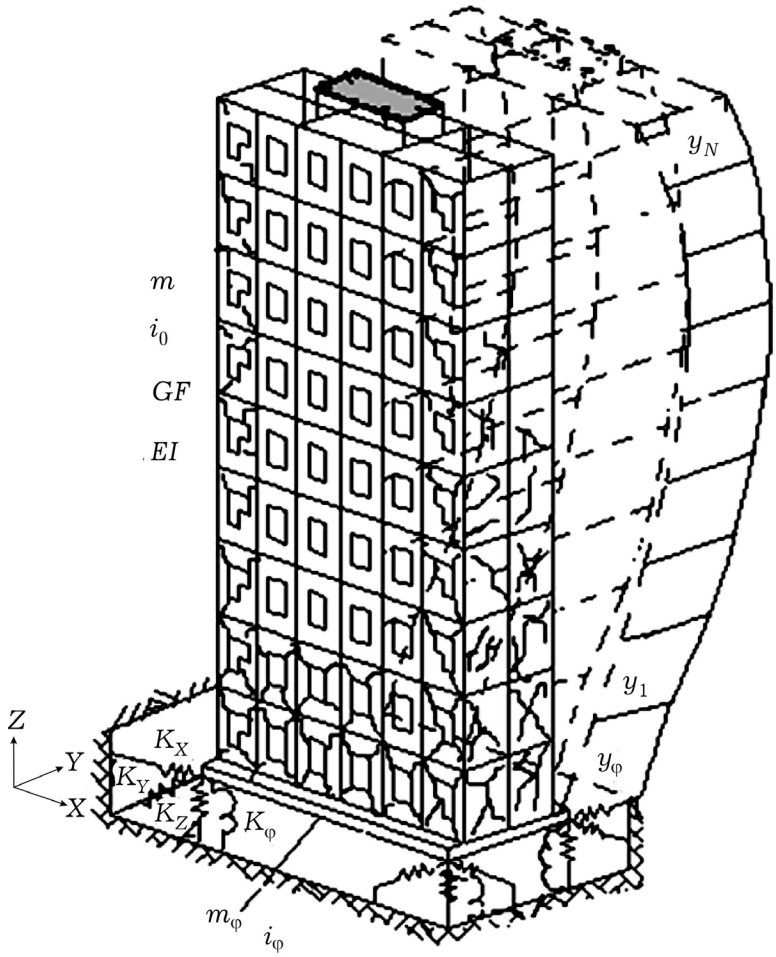


Fig. 3. Spatial CDM of the building taking into account the base and seismic isolation stiffnesses (K_x, K_y, K_z, K), foundation mass m_φ and moment of inertia i_φ, masses of the stories structures m and moments of inertia i_0, shear GF and bending EI stiffness of stories structures.

$$
\begin{aligned}
&m_1\ddot{x}_1 + (K_x + K_{12})\,x_1 - K_{12}x_2 + H\,\operatorname{sgn}\dot{x}_1 + K_{12}h_{12}\varphi = -m_1\ddot{x}_0,\\
&m_2\ddot{x}_2 - K_{12}x_1 + (K_{12} + K_{23})\,x_2 - K_{23}x_3 + (K_{23}h_{23} - K_{12}h_{12})\,\varphi = -m_2\ddot{x}_0,\\
&m_3\ddot{x}_3 - K_{23}x_2 + (K_{23} + K_{34})\,x_3 - K_{34}x_4 + (K_{34}h_{34} - K_{23}h_{23})\,\varphi = -m_3\ddot{x}_0,\\
&m_4\ddot{x}_4 - K_{34}x_3 + (K_{34} + K_{45})x_4 - K_{45}x_5\\
&\qquad + (K_{45}h_{45} - K_{34}h_{34})\,\varphi = -m_4\ddot{x}_0,\\
&m_5\ddot{x}_5 - K_{45}x_4 + K_{45}x_5 - K_{45}h_{45}\varphi = -m_5\ddot{x}_0,\\
&i_{\text{total}}\ddot{\varphi} + K_{12}h_{12}x_1 + (K_{23}h_{23} - K_{12}h_{12})\,x_2
\end{aligned}
\tag{2}
$$

$$+ (K_{34}h_{34} - K_{23}h_{23})\,x_3 + (K_{45}h_{45} - K_{34}h_{34})\,x_4$$

$$- K_{45}h_{45}x_5 + \left(K_{12}h_{12}^2 + K_{23}h_{23}^2 + K_{34}h_{34}^2\right.$$

$$\left. + K_{45}h_{45}^2 + K_{\varphi} + K_{\varphi}^{12}\right)\varphi + \left(K_{\varphi}^{23} - K_{\varphi}^{12}\right)\vartheta_2$$

$$+ \left(K_{\varphi}^{34} - K_{\varphi}^{23}\right)\vartheta_3 + \left(K_{\varphi}^{45} - K_{\varphi}^{34}\right)\vartheta_4 - K_{\varphi}^{45}\vartheta_5 = 0,$$

$$i_2\ddot{\vartheta}_2 + \left(K_{\varphi}^{23} - K_{\varphi}^{12}\right)\varphi + \left(K_{\varphi}^{12} + K_{\varphi}^{23}\right)\vartheta_2 - K_{\varphi}^{23}\vartheta_3 = 0,$$

$$i_3\ddot{\vartheta}_3 + \left(K_{\varphi}^{34} - K_{\varphi}^{23}\right)\varphi - K_{\varphi}^{23}\vartheta_2 + \left(K_{\varphi}^{23} + K_{\varphi}^{34}\right)\vartheta_3 - K_{\varphi}^{34}\vartheta_4 = 0,$$

$$i_4\ddot{\vartheta}_4 + \left(K_{\varphi}^{45} - K_{\varphi}^{34}\right)\varphi - K_{\varphi}^{34}\vartheta_4 + K_{\varphi}^{45}\vartheta_4 - K_{\varphi}^{45}\vartheta_5 = 0,$$

$$i_5\ddot{\vartheta}_5 - K_{\varphi}^{45}\varphi - K_{\varphi}^{45}\vartheta_4 + K_{\varphi}^{45}\vartheta_5 = 0.$$

System (2) of the governing differential equations was solved numerically in the Mathcad environment by using the Runge–Kutta method.

2. Equation of Motion for the Spatial Digital Twin of a Separate Building under the Dynamic Effects of Railway Transport

The numerical studies of the forced vibrations of spatial multimass linear CDM of buildings (Fig. 3) under the effects of railway transport are performed by using the LIRA CAD software package (SP) in which the finite-element method is implemented [19]. In the LIRA CAD SP, the finite-element spatial model of the building is calculated for the railway transport dynamic effects (of low intensity, at which building structures work in an elastic phase) by using the following system of differential equations with constant coefficients:

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{K}(\mathbf{x})\mathbf{x}(t) = -\mathbf{M}\ddot{\mathbf{x}}_0(t). \tag{3}$$

Here, $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ are the constant matrices of the system masses, damping and stiffness; $\ddot{\mathbf{x}}(t)$, $\dot{\mathbf{x}}(t)$, and $\mathbf{x}(t)$ are the accelerations, velocities and displacements vectors; $\ddot{\mathbf{x}}_0(t)$ is the kinematic load (soil acceleration at the building base) corresponding to time t.

The initial velocities are assumed to be zero, and the initial displacements are obtained from the solution of the system of equations for the first loading $\mathbf{x}(t) = \mathbf{x}_1$.

If (3) is considered as a system of ordinary differential equations with constant coefficients, the velocities, accelerations, and displacements can be approximated by the finite difference expressions for displacements. Using the central differences method, the system of equations of motion for accelerations at time t can be written in the following form:

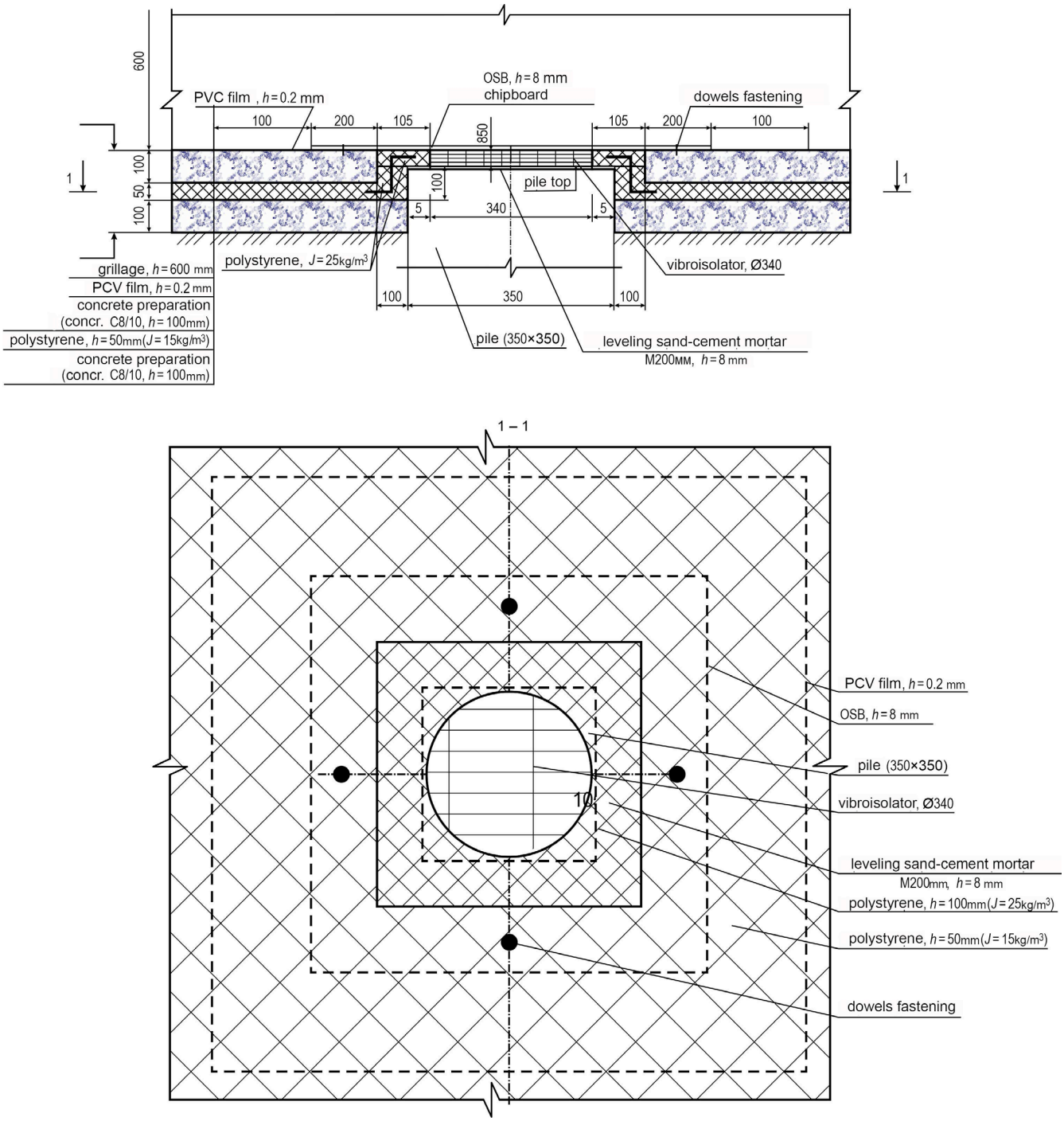


Fig. 4. Schematic diagram of the vibration and seismic protection system of the building with rubber supports (vibration isolators with a diameter of 340 mm and a height of 50 mm) on the pile-grid level: PVC film, OSB (oriented strand board) fasten with dowels, pile top, Styrofoam (kg/m^3), vibration isolator, grid, concrete bed (concrete of C8/10 grade, $h = 100$ mm), pile, leveling sand grout.

$$\left(\frac{2\mathbf{M}}{(\Delta t)^2} + \frac{\mathbf{C}}{\Delta t} + \mathbf{K}\right)(\mathbf{x}(t+\Delta t) + \mathbf{x}(t-\Delta t))$$

$$= 2\left(-\mathbf{M}\ddot{\mathbf{x}}_0(t) + \frac{2\mathbf{M}}{(\Delta t)^2}\mathbf{x}(t) + \frac{\mathbf{C}}{\Delta t}\mathbf{x}(t-\Delta t)\right). \quad (4)$$

The "new" displacements $\mathbf{x}(t+\Delta t)$ are determined taking into account the displacements $\mathbf{x}(t)$ and $\mathbf{x}(t-\Delta t)$ previously obtained when solving system (4). Such integration schemes are called implicit integration schemes. The given integration scheme is called the modified method of central differences. Equations (4) are initial for solving both linear and nonlinear problems by direct dynamic calculations in the LIRA CAD software package [34].

In the numerical studies, the options are considered with allowance for the dry friction forces during the vibrations of the reinforced-concrete foundation on the rubber supports relative to the base made of polystyrene foam (Fig. 4).

3. Experimental Studies of the Soil and Buildings Vibration under the Effects Caused by Railway Trains

The construction site is in a densely built-up area with neighboring highways and railways. Traffic near the construction site is unlimited during the day. The two-way traffic of passenger and freight trains through this section of Kyiv–Lviv railway located near the construction site is also not restricted in any way.

In compliance with the design, the PDRC consists of three sections differing by the stories number and dimensions in plan (Fig. 1). Section 1 is located at a minimum distance from the railway track (20–25 m). The railway track is ordinary one without any vibration isolation elements and with butt joints. Prior to the building construction start, the soil surface vibrations at the residential complex construction site were experimentally studied. A comparison of the experimental and allowable values of the levels of vibration accelerations showed that under the railway trains effects the soil vibrations levels and predicted floors vibrations levels exceeded the values allowable for residential buildings by 6–12 dB (from two to four times). Thus, the necessity was proved to install a vibration protection system at the pile grid level allowing to reduce the structures vibrations levels and ensure the comfortable living conditions in buildings.

The soil accelerations records obtained at the residential complex construction site are used in the dynamic analysis of the three sections spatial models for determining the predicted levels of building floors vibrations accelerations. The numbers of stories in sections are as follows: 13, 6, and 10 in sections 3, 2, and 1, respectively. Each section is arranged on its own vibration-isolated pile foundation. The frame of section 3 is made girderless of cast-in-place reinforced concrete with the stiffening cores in the zone of an elevator and stair blocks. In plan, the building floors have variable areas with decreasing height. The floor plan has the shape of a rectangular trapezoid. The frame of section 2 is girderless of cast-in-place reinforced concrete with the stiffening cores in the elevator block zone. In plan, the section has a trapezoidal shape and adjoins sections 3 and 1. The section 1 frame is girderless of cast-in-place reinforced concrete with the stiffening cores in the elevator and stair blocks zone. The floor shape of section 1 is close to a parallelogram in plan.

According to [3], the construction site design seismicity is 7 points. Therefore, the construction should ensure the protection against both the ground railway transport dynamic effects and the earthquakes impacts.

The vibrometric studies of the soil, vibration-isolated grids and floor slabs were carried out over a period of two years during the construction of a residential complex with a system of vibration and seismic protection against railway trains effects and earthquakes.

The state of construction of the residential complex at the time of vibrometric studies carried out in November 2018 is shown in Fig. 5.



Fig. 5. General view of the construction site on 22.11.2018 after erection of all stories in three sections of the residential complex.



Fig. 6. The vibration and seismic protection system of the buildings arranged at the pile grid level (Fig. 2).

The final vibrometric studies and concrete grades determination by non-destructive control methods were conducted at the object in November 2018 after erection of all stories in three sections (6, 10, and 13 stories). The vibrodynamic effects on the adjacent soil, grid (Fig. 6) and buildings floors were studied without moving railway trains (microseismic vibrations and natural seismicity of the Vrancea zone) and with them (man-made dynamic effects from railway trains and motor vehicle traffic). The dynamic studies were organized and conducted in compliance with the regulatory documents [2, 21]. The vibrometric equipment allowing recording the

vibration signals in real time and carrying out their processing with the octave and narrow-band spectra determination was used during the experimental studies. The obtained vibration signals (vibrations accelerations) were recorded and processed by means of the Seismomonitoring software package [13].

Based on the performed studies, the vibrations accelerations instrumental records were obtained in real time during the vibrations of the adjacent soil and structures of residential complex sections 1, 2, and 3 in the 0.3–100 Hz frequency range under the railway trains dynamic effects.

As a result of the obtained records processing by the spectral analysis method, the levels of dynamic effects on the adjacent soil and structures of three building sections were determined, and the vibrations frequencies and maximum vibrations levels were recorded. In the 13- and 6-storey sections the recorded maximum levels of the floors vertical vibration accelerations in the 16 Hz octave band reached 68 dB (Fig. 7), which was less than the 71 dB value allowable according to the Sanitary Standards [4].

Based on the comparison of the recorded levels of vibration effects on three sections structures in the residential complex and the values allowable according to regulatory documents, a conclusion was made as to ensuring the allowable values of vibrations levels under the moving trains dynamic effects.

4. Analysis of the Dynamic and Seismic Effects

The structural scheme of the residential 6, 10, and 13-storey buildings is a girderless cast-in-place reinforced-concrete frame (Fig. 1). The load-bearing vertical structures are composed of columns, pylons, and stiffening cores. The typical storey height is 3.0 m. The storey floors and coverings are 200-mm-thick slabs of cast-in-place reinforced concrete connecting the vertical elements and ensuring the spatial stiffness of the building. The buildings foundations are cast-in-place reinforced-concrete grids on a pile basis. The accepted piles cross section is $350\ \text{mm} \times 350\ \text{mm}$ in size.

The pile grids are designed with a thickness of 600 mm in the 6-storey building and a thickness of 800 mm in the 10- and 13-storey buildings. For protecting against the railway trains dynamic effects, a vibration and seismic protection system is arranged (Figs. 4 and 6) based on the results of numerical studies and tests of rubber vibration isolators (Fig. 8). The rubber isolator with 340 mm diameter and 50 mm or 40 mm thickness (Fig. 6) is installed on each pile head before concreting the grid plate. The proposed solution of the vibration protection system was also tested in 10- and 27-storey residential buildings in Kyiv. The floors vibration levels in twelve erected buildings did not exceed the values allowable under the railway trains dynamic effects (shallow and deep subway lines) according to the Sanitary Standards [6].

The spatial digital twins vibrations are calculated for the train dynamic effects in three sections of the residential complex with and without the vibration protection system (rubber vibration supports, on which the grid rests) using the LIRA CAD software package [23], in which the finite element method is implemented. The experimental nonlinear relationships of vertical displacements and loads of isolators are shown in Fig. 8.

The vertical vibration frequencies in the 13- and 10-storey sections built on vibration supports (isolators with compressive stiffness $K_z = 105000$ kN/m) are equal to 3.8 Hz and 4.1 Hz, respectively. The vertical vibration frequency in the 6-storey building section built on vibration supports (isolators with compressive stiffness $K_z = 67000$ kN/m) is as high as 4.7 Hz.

For the vertical vibrations of floors in the 6-storey building, the design levels of 0.00477 m are obtained in the absence of vibration protection (Fig. 9), while the design levels of 0.000724 m are obtained with the installed vibration protection system (Fig. 10), i.e., it is possible to predict a 6.6-fold weakening of vibrations of the floors. The same data are obtained when calculating the digital twins of the 10- and 13-storey sections.

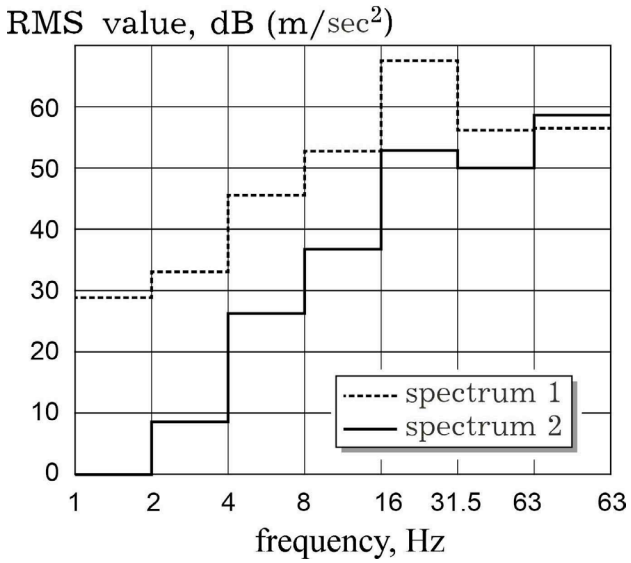


Fig. 7. Octave spectra of the vertical vibration accelerations during the freight train passage (vibration sensors 1 and 2 are installed on the 2nd-storey floor in section 3 and on the 2nd-storey floor in section 2, respectively): spectrum 1, spectrum 2, the RMS value, dB (m/sec^2), and frequency (Hz).

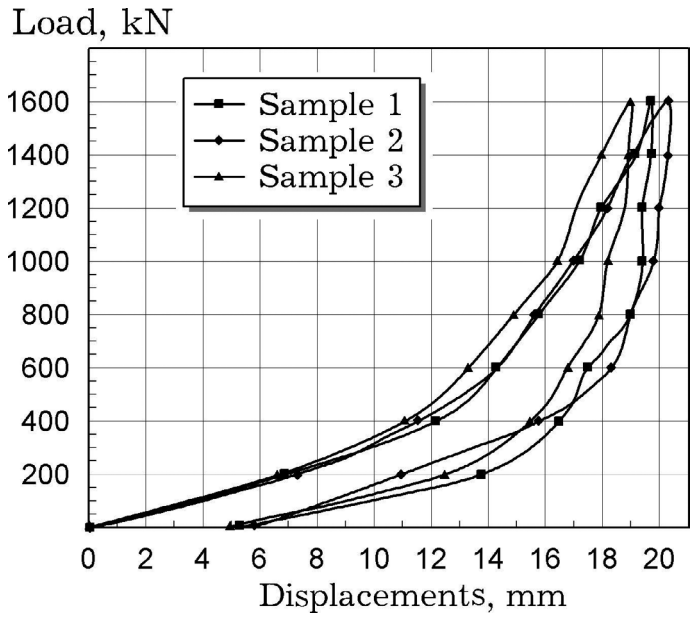


Fig. 8. The experimental nonlinear dependences of vertical displacements on the loads (of up to 1600 kN) for the rubber isolators 340 mm in diameter and 50 mm in height: samples *1*, *2*, and 3; load (kN), and displacements (mm).

The safety factor K_s for buildings overturning under seismic and wind loads is determined by the following formula: $K_s = M_m / M_s$, where M_m is the minimum resisting moment against the permanent load with respect to the edge row of vibration isolators, M_s is the maximum overturning moment for seismic or wind loads.

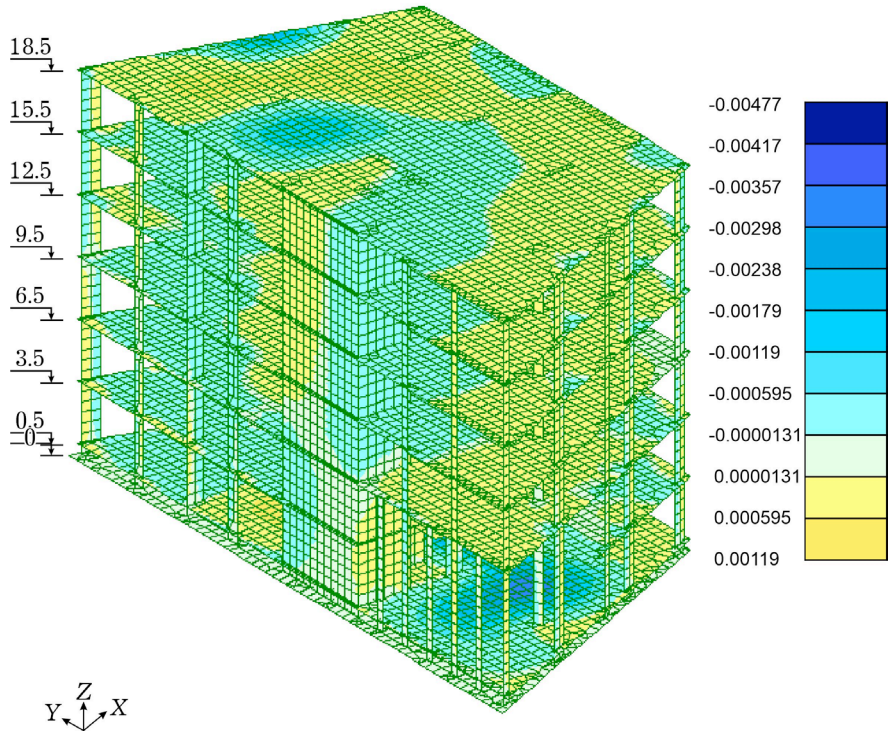


Fig. 9. The design vertical amplitudes for the displacements of the 6-storey-building floors under the actions of passenger trains (the 55th form of vibration) in the absence of vibration protection systems: SEISMICS 2, component 55, isofields of displacements along $Z(G)$, measurement units (mm), the masses are collected for loads 1 and 2.

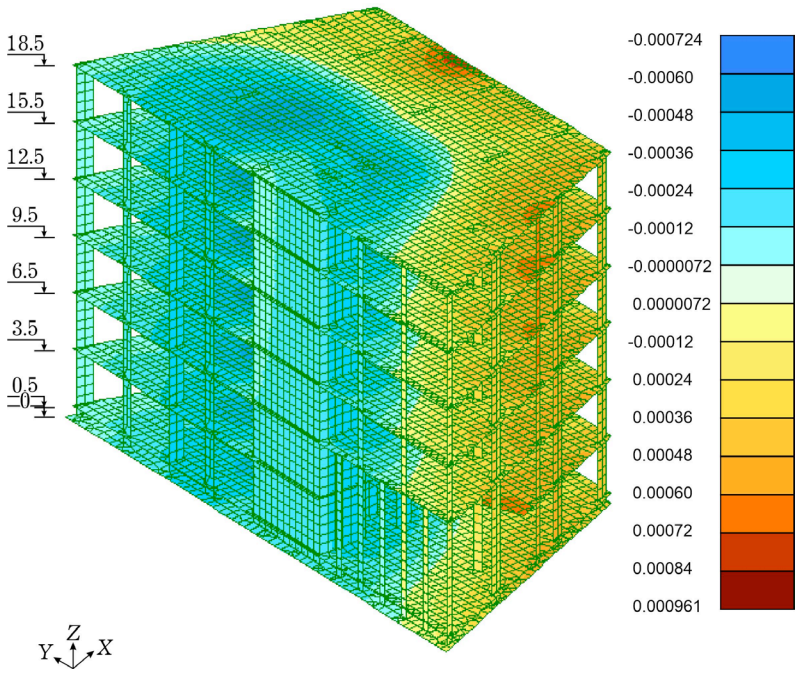


Fig. 10. The design vertical amplitudes of displacements for the floors under the actions of passenger train (the 10th form of vibrations) in the 6-storey building equipped with a vibration protection system: SEISMICS 2, component 10, isofields of displacements along $Z(G)$, measurement units (mm), and the masses are collected from loads 1 and 2.

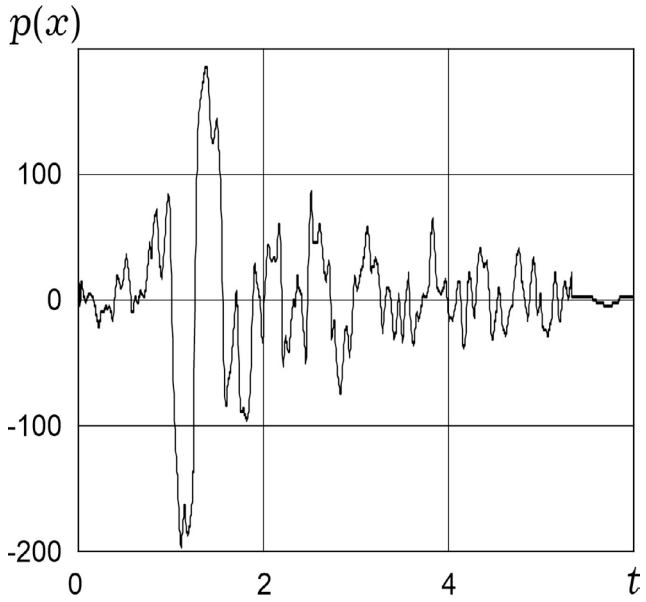


Fig. 11. Accelerogram of the long-period Bucharest earthquake (1977) of the 8 points intensity (the maximum ground acceleration $p = 2$ m/sec^2 was recorded at the beginning of the earthquake; time $t = 1.0$ sec) [24].

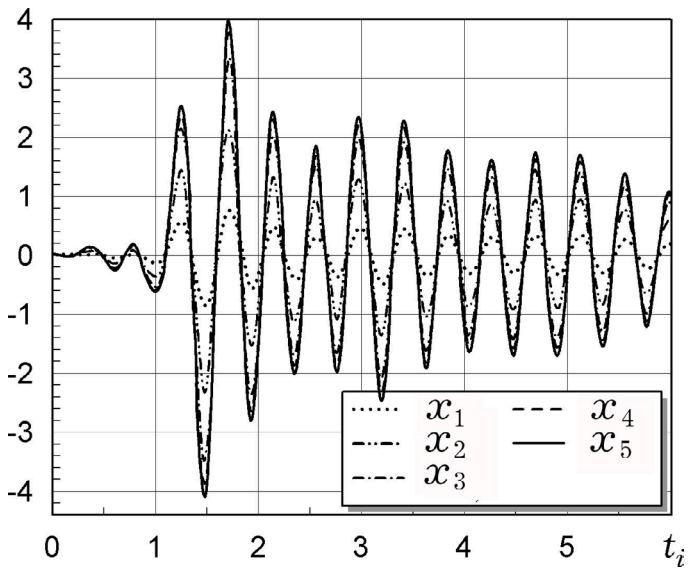


Fig. 12. The horizontal displacements $(\text{m}\times 10^{-2})$ of the building foundation (x_1) and stories $(x_2,\ldots,x_5)$ as functions of time t (sec) caused by the impact of the 1977 Bucharest earthquake with a magnitude of 8 points (the frictional force is assumed to be as high as 0.8% of the building weight).

The calculations of overturning safety factors for the 13-storey building (the maximum height among all three analyzed sections of the residential complex) confirm that for seismic effects with a magnitude of 7 points the minimum design safety factor should be not smaller than 5.4. For the wind effects, the minimum safety factor should be at least 101.6.

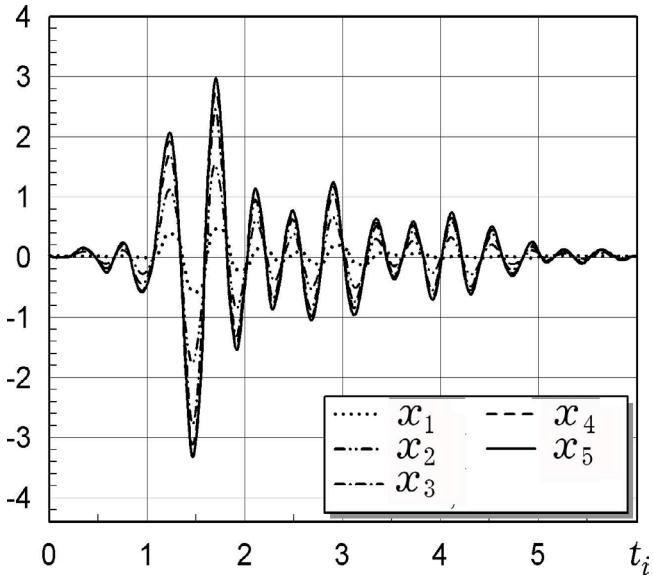


Fig. 13. The horizontal displacements $(\mathrm{m}\times 10^{-2})$ of the building foundation (x_1) and stories $(x_2,\ldots,x_5)$ at time t (sec) under the impact of the 1977 Bucharest earthquake with a magnitude of 8 points (the friction force is estimated as 8% of the building weight).

5. Results of the Nonlinear Flat Model Analysis of Seismic Effects upon the Building

Earthquakes occurring in the Vrancea area (Romania) prove to be most dangerous for the city of Lviv [3]. Therefore, the nonlinear flat model (Fig. 2) is analyzed for the effects of the accelerogram (Fig. 11) recorded in the course of the devastating Bucharest earthquake in 1977. The numerical studies of the nonlinear equations (1) were carried out in the Mathcad environment. The building foundation was assumed to have a mass of 1200 tons, while the mass of stories was assumed to be equal to 500 tons. The analysis of the results show (see Figs. 12 and 13) that it is possible to additionally decrease the displacements of the building foundation during earthquakes by a factor of 1.5 if the friction forces are taken into account.

CONCLUSIONS

The theoretical and experimental dynamic surveys of the buildings complex located at 26 Pid Dubom St. in Lviv based on the technology of digital twins included the experimental investigations of soil surfaces in the construction site as well as of the grids and floors structures of the buildings by using the IoT; the numerical studies of the spatial digital twins of 6-, 10- and 13-storey buildings with and without vibration protection systems for the dynamic effects caused by railway trains, and the numerical studies of the flat digital twins of 6-, 10-, and 13-storey buildings for the seismic effects of the Vrancea zone.

Based on the results of the theoretical and experimental dynamic studies, we can make the following conclusions:

1. The comparison of the experimental and admissible values of the levels of vibration accelerations shows that, under the effects of railway trains, the soil vibration levels exceed the values admissible for residential buildings by $6-12$ dB (2–4 times). This proves the necessity of creation of vibration protection system on the

pile-grid level aimed at lowering the levels of vibration in structures and guaranteeing comfortable living conditions in the buildings.

2. The numerical analysis of the spatial digital twins of three residential sections of the complex was carried out under the actions of railway trains with and without vibration protection systems.

3. To ensure comfortable living conditions in the buildings of residential complex, we developed and constructed a vibration and seismic protection system at the pile foundation level. The designed frequency of natural vertical vibrations of the buildings with vibration supports constitutes 3.8–4.7 Hz, which is (3.12 times) lower than the frequencies of soil-forced vibrations (15–80 Hz) caused by the railway-trains effects. The design and experimental data analyses show that, in the presence of installed vibration isolation, the vibration levels of the floors do not exceed the levels permitted by the Sanitary Standards for residential buildings. In the absence of vibration protection, the design levels of the vertical vibrations of floors are 1.5–4 times higher than their admissible values (from 2.9 dB to 13.0 dB).

4. The proposed system of vibration and seismic protection of the building realized on the pile foundation level ensures both the possibility of damping in the rubber supports (up to 5% of the critical damping) and the presence of friction forces, which allows one to additionally decrease the displacements of the building foundation during earthquakes by a factor of 1.5.

5. The minimum design factor of the safety of buildings against overturning is equal to 5.4 in the case of seismic loads (determined for 7-point magnitudes) and to 6.5 under the wind effects.

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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